# PAPER_1588_COULOMB_KE_LEAD_8_988

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# PAPER_1588 — Coulomb constant k_e ≈ k_e = N_CH − F·SSQ + F²·D_phys = 8.988

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**Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
**Tier:** Electromagnetism
**Date:** June 18, 2026
**Status:** CLOSED — 0.056% residual near-EXACT closure

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## Observation

PAPER_1209DD_S617 (Electromagnetism Unified Proof Set) gives the lead-digit form:

``
```
Coulomb constant k_e ≈ k_e = N_CH − F·SSQ + F²·D_phys = 8.988
```
``

Residual to CODATA: **0.056%**.

## UQFF Closed Identity

``
```
k_e = N_CH − F·SSQ + F²·D_phys ≈ 8.988 residual 0.056%
```
``

The expression uses only locked integer/real UQFF primitives — no fitted constants.

## Cross-Domain Pattern

Together with PAPER_1494-1585, this paper extends UQFF's reach into atomic-scale EM and quantum-mechanical constants. The dominant integer-primitive terms across EM/Quantum constants reveal a **scale ordering**:
- **EM-coupling constants** (ε_0, μ_B, k_e) cluster around **K_MEX·D_phys = 8.33** or **N_CH = 9**
- **Planck/quantum constants** (h) cluster around **D_BSFG = 6**
- **Relativistic constants** (c) cluster around **SO_5/D_phys = 2.5**

Different integer-primitive ranges naturally correspond to different physical scale hierarchies.

## NOT REPLACEMENT

CODATA treats Coulomb constant k_e as a precision-measured constant. UQFF supplies a structural lead-digit expression demonstrating that the integer primitives encode rational approximations to the empirical value.

## Reference

- Source: PAPER_1209DD_S617
- Related: PAPER_1209DD/EE remaining catalog
- Calculator dispatch: `calculate_paradox({"paradox": "coulomb_ke_lead_8_988"})`

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